## Experiment No 8

**Aim: To develop a Risk Mitigation, Monitoring and Management (RMMM) Plan for Project**

RMMM stands for Risk Mitigation, Monitoring and Management. It is used in software engineering to identify possible risks in a project and define actions to:

- Mitigate the risk before it occurs.
- Monitor the risk during the project.
- Manage the risk if it actually occurs.

For the Library Management System, RMMM helps reduce problems related to requirements, technology, data, security, schedule, cost, and users.

**RMMM Plan**

RMMM Assessment identifies the probability and impact of each risk and helps determine its priority.
RMMM Assessment Table – Library Management System

| Risk | Probability | Impact | Risk Exposure (P × I) | Risk Level | Mitigation | Monitoring | Management |
|---|---|---|---|---|---|---|---|
| Requirement changes | Medium (3) | High (4) | 12 | High | Clearly define requirements and get user approval | Review requirements regularly | Update SRS and modify project plan |
| Loss of database/data | Low (2) | Very High (5) | 10 | High | Take regular database backups | Check backup status | Restore data from latest backup |
| Security breach | Medium (3) | High (4) | 12 | High | Use authentication, authorization and encryption | Monitor login attempts and security logs | Block unauthorized access and recover affected data |
| Software failure/bugs | Medium (3) | High (4) | 12 | High | Perform unit, integration and system testing | Track defects continuously | Fix bugs and perform regression testing |
| Server/system failure | Low (2) | High (4) | 8 | Medium | Maintain backup server/system and recovery procedures | Monitor system performance | Restore service using backup/recovery system |
| Project delay | Medium (3) | High (4) | 12 | High | Prepare Gantt chart and realistic schedule | Compare actual progress with planned progress | Reallocate resources and revise schedule |
| Lack of technical skills | Medium (3) | Medium (3) | 9 | Medium | Provide training and documentation | Monitor team performance | Assign experienced members or provide additional training |
| User resistance | Medium (3) | Medium (3) | 9 | Medium | Provide user training and simple interface | Collect user feedback | Improve interface and conduct additional training |
| Incorrect book/member data | Medium (3) | High (4) | 12 | High | Use validation checks and proper data-entry rules | Perform regular data audits | Correct invalid records and restore from backup if required |
| Internet/network failure | Medium (3) | Medium (3) | 9 | Medium | Provide reliable network and offline procedures where possible | Monitor network availability | Restore network or use backup connectivity |
| Integration problems | Low (2) | Medium (3) | 6 | Medium | Define interfaces clearly and test integration early | Perform integration testing | Isolate and fix interface-related defects |
| Insufficient resources | Medium (3) | Medium (3) | 9 | Medium | Estimate hardware, software and manpower requirements | Review resource usage | Reallocate or obtain additional resources |

**Risk Rating Scale**

**Probability:**

- 1 = Very Low

- 2 = Low
- 3 = Medium
- 4 = High
- 5 = Very High

**Impact:**
- 1 = Very Low
- 2 = Low
- 3 = Medium
- 4 = High
- 5 = Very High

**Risk Exposure** = Probability × Impact

**Risk Priority**

| Risk Exposure | Priority |
|---|---|
| 1–4 | Low |
| 5–9 | Medium |
| 10–15 | High |
| 16–25 | Very High |

**Conclusion:**
For the Library Management System, the major risks requiring close attention are requirement changes, security breaches, software failure, project delays, database/data loss, and incorrect data.